Knowledge Graph Lab

KG Lab WS 2025

Content

- People
- Motivation
- What is the KG Lab?
- What will we do?
- What are the goals?
- What are the prerequisites?
- How will you achieve the necessary knowledge to start?
- Summary
- Questions/Discussion
- Next steps

People

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People

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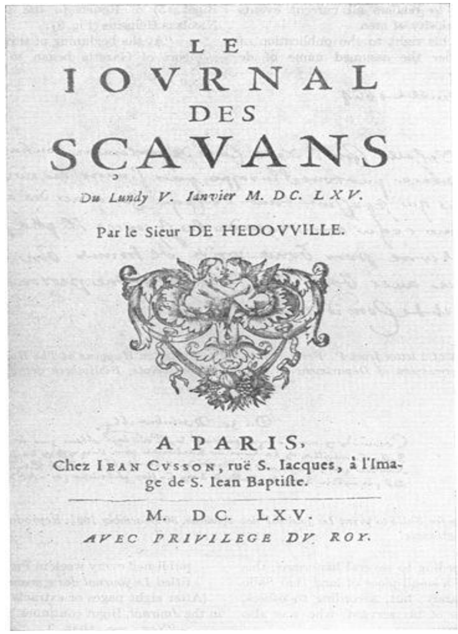
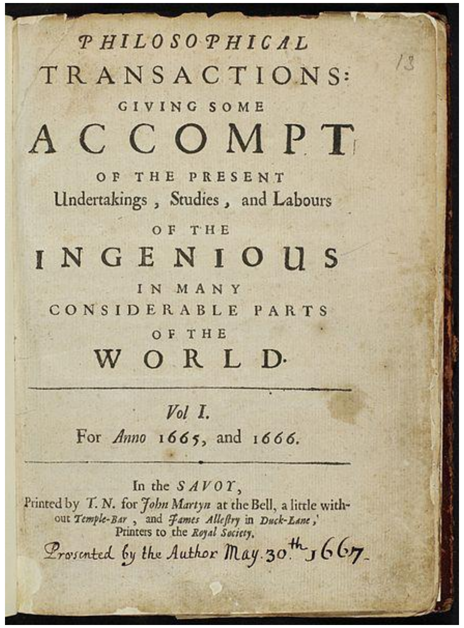
Dissemination of Knowledge 1665… until the 70s



Journal des Sçavans



"Philosophical Transactions of the Royal Society"

"Philosophical Transactions of the Royal Society"

The Growth of Knowledge….

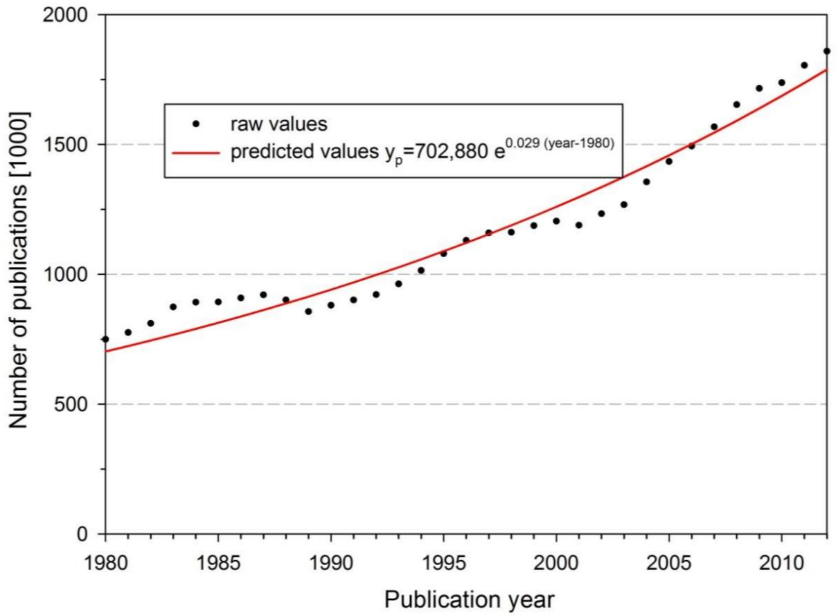


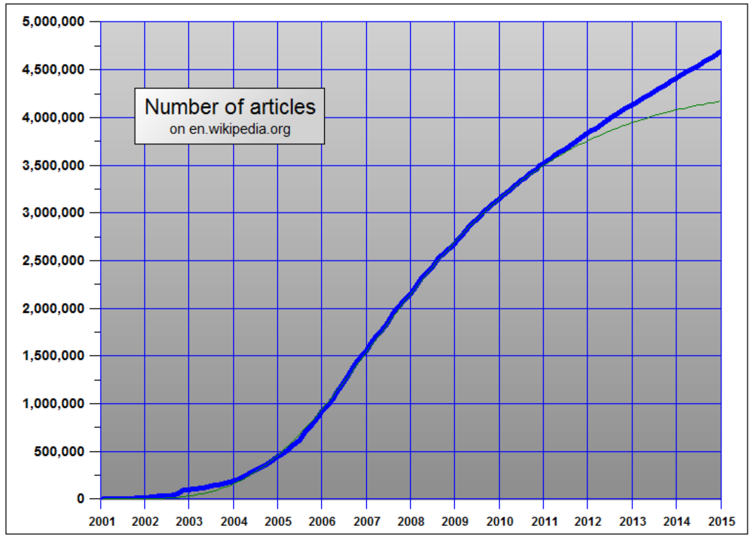
Figure 1. Exponential growth of scientific output from 1980 to 2012

Bornmann, Lutz, and Rüdiger Mutz. "Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references." Journal of the Association for Information Science and Technology 66.11 (2015): 2215-2222.

- Growth of 8-9% per year
- Doubling of Knowledge every 9 years

"The difficulty seems to be […] that publication has been extended far beyond our present ability to make real use of the record."

Vannevar Bush: As We May Think. Atlantic Monthly, 176, 101-108, July 1945.



Growth of knowledge in Wikipedia

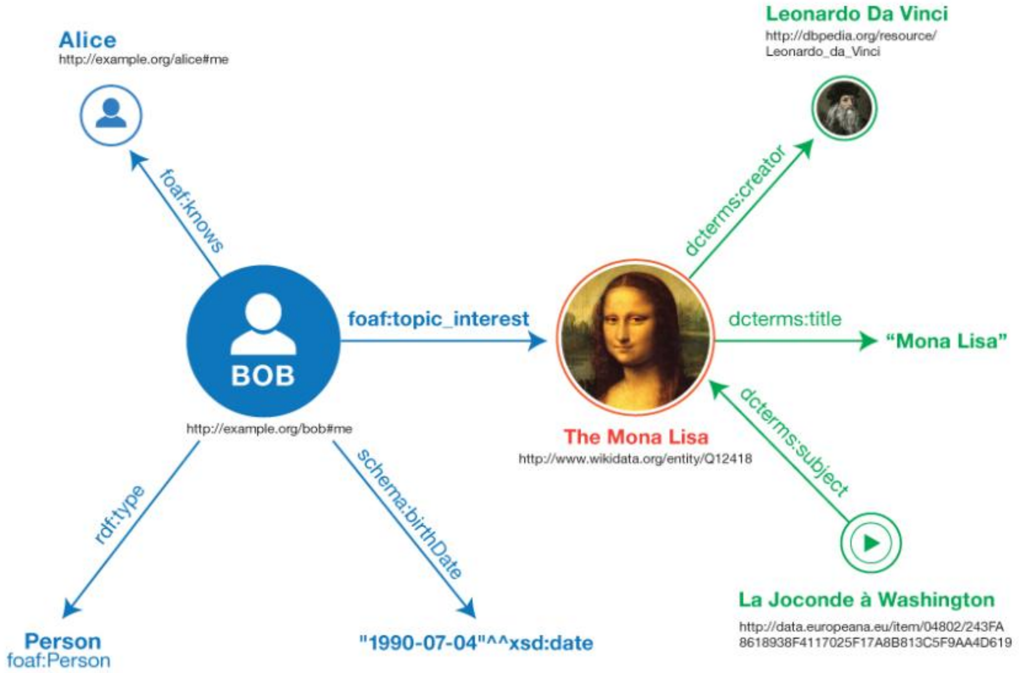
About Data and Knowledge Silos

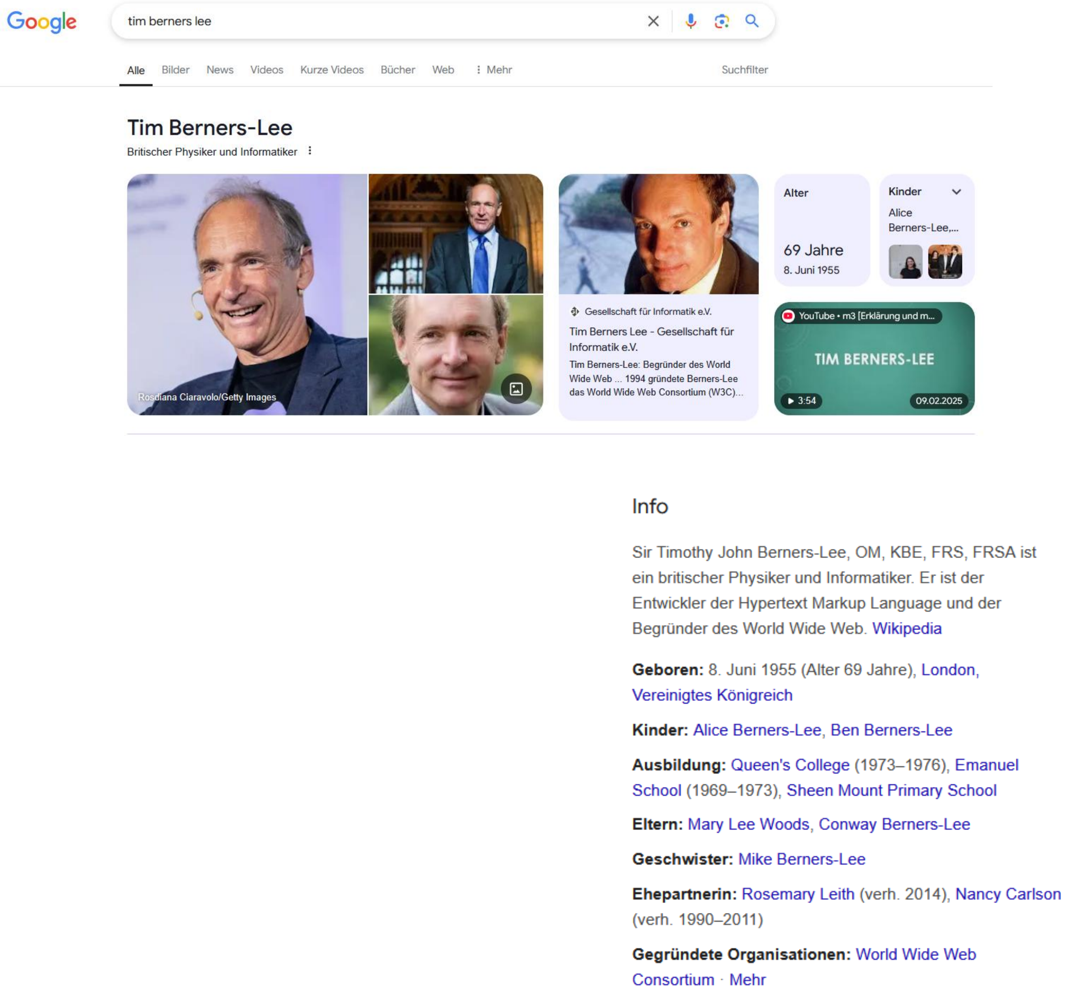
- Every result is separated from all others

Motivation

Why Knowledge Graphs?

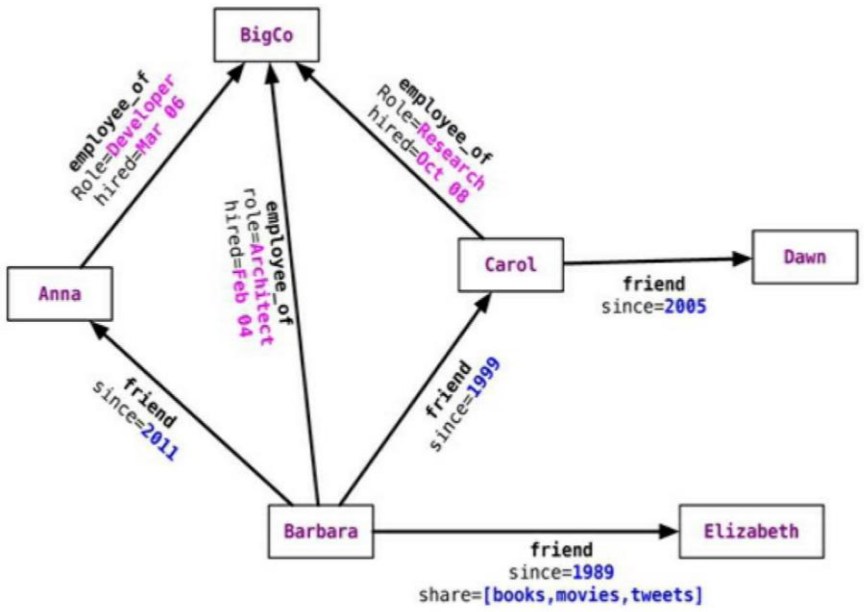
We use knowledge graphs all the time …
… but we might not recognize the knowledge graphs





What is a Graph?

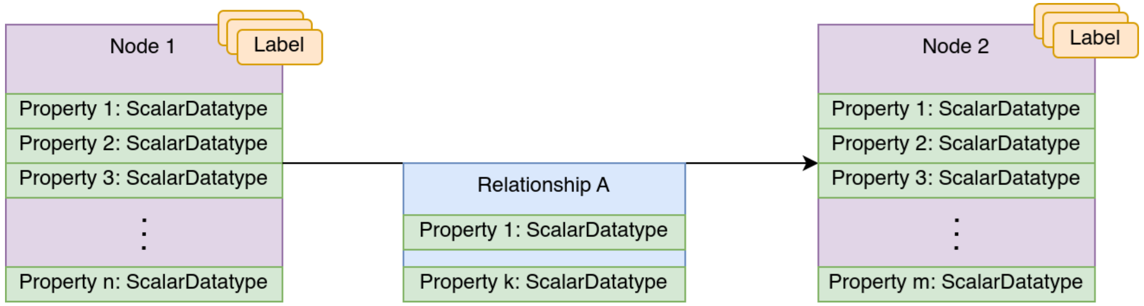
- Collection of nodes and edges
- Nodes and edges can have properties
  - indicate what entities they represent
- Edges can have directions (directed/undirected graph)
- Queries traverse graph structure using path expressions
- allow dynamic schema evolution



Data Structure

- RDF-Graph
  - RDF (*Resource Description Framework*)
  - consists of triples
  - three kinds of nodes:
    - IRIs
    - literals
    - blank nodes
  - Query language: SPARQL

- Property Graph
  - consists of nodes, edges, properties, labels
  - Nodes and edges can have properties
    - of scalar data types
  - Query language: Cypher



Knowledge Graph

Definitions

“Knowledge Graphs are a very large semantic nets that integrate various and heterogeneous information sources to represent knowledge about certain domains of discourse” [Fensel et al.]

“Knowledge graph as a data graph potentially enhanced with representations of schema, identity, context, ontologies and/or rules. These additional representations may be embedded in the data graph, or layered above it.” [Hogen et al.]

Semantic descriptions of entities and their relationships:

- Uses a knowledge representation formalism
- **Entities**:
  - real world objects (things, places, people) and
  - abstract concepts (genres, religions, professions)
- **Relationships**: graph-based data model where relationships are first-class

Knowledge Graph History

- 2007: DBpedia and Freebase
  - based on Wikipedia
- 1. Jan 2012 Google Knowledge Graph
  - includes RDFa, JSON-LD and microdata from crawled webpages
  - based on Freebase and DBpedia
  - ontology organized over schema.org
- 29. Oct 2012 Wikidata
  - Integrated Freebase
- 2015 Microsoft Bing KG
- 2019 Inception of The Knowledge Graph Conference
- 2020 The EU Knowledge Graph
- 2024 dblp Knowledge Graph (dblp KG)

KG – Maintenance

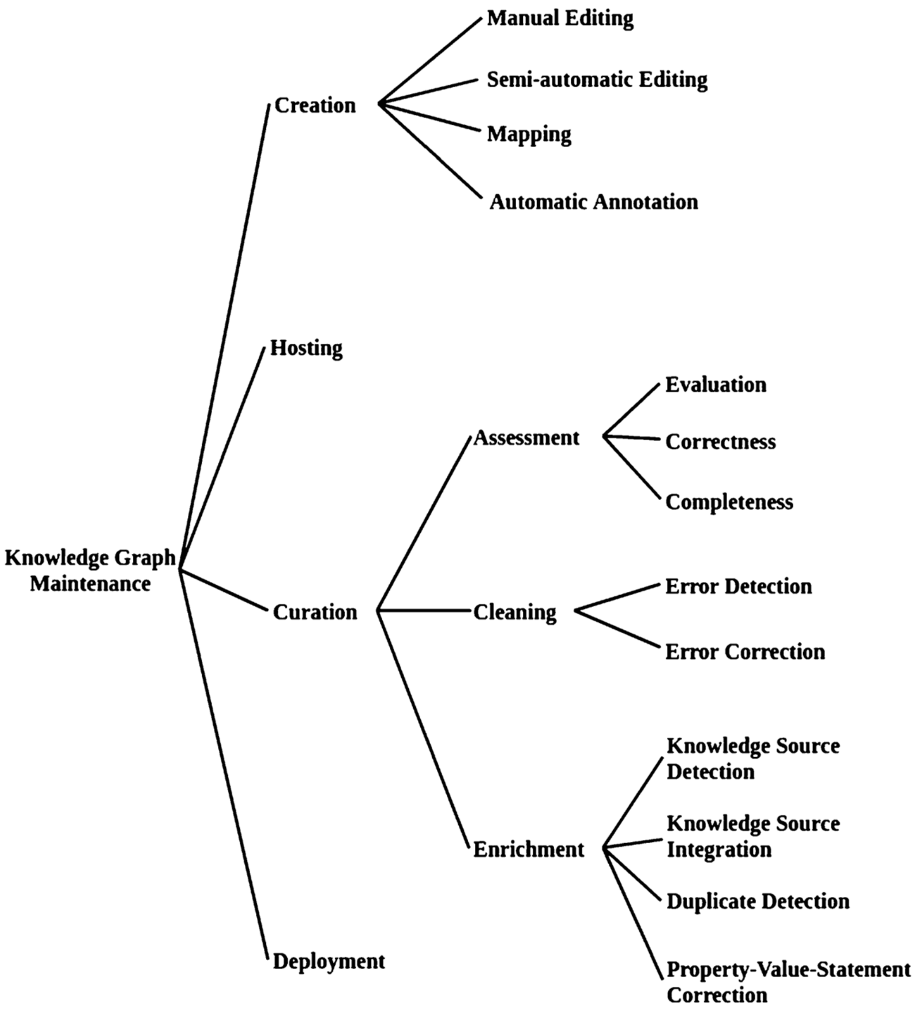
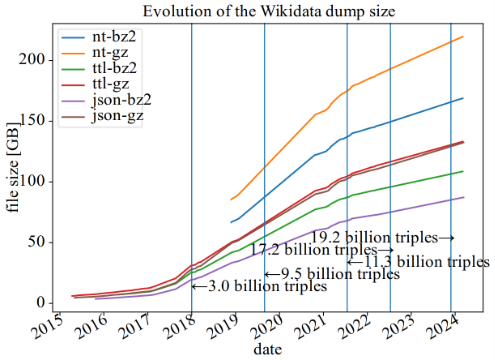
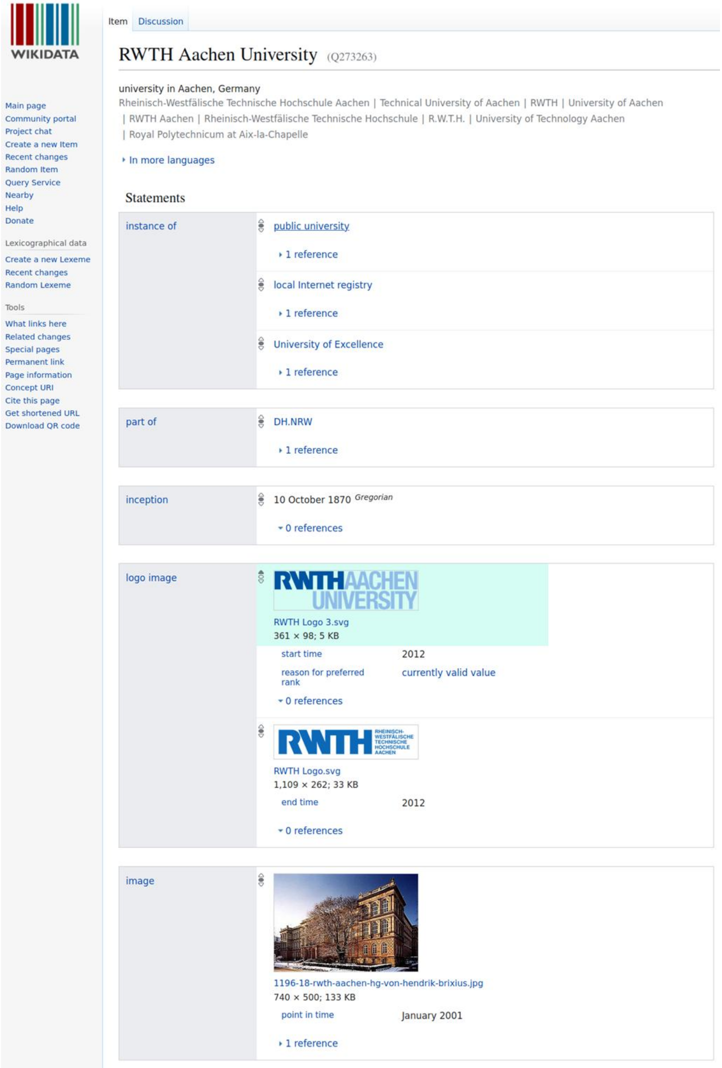


Fig. 2.2 A task model for Knowledge Graph generation

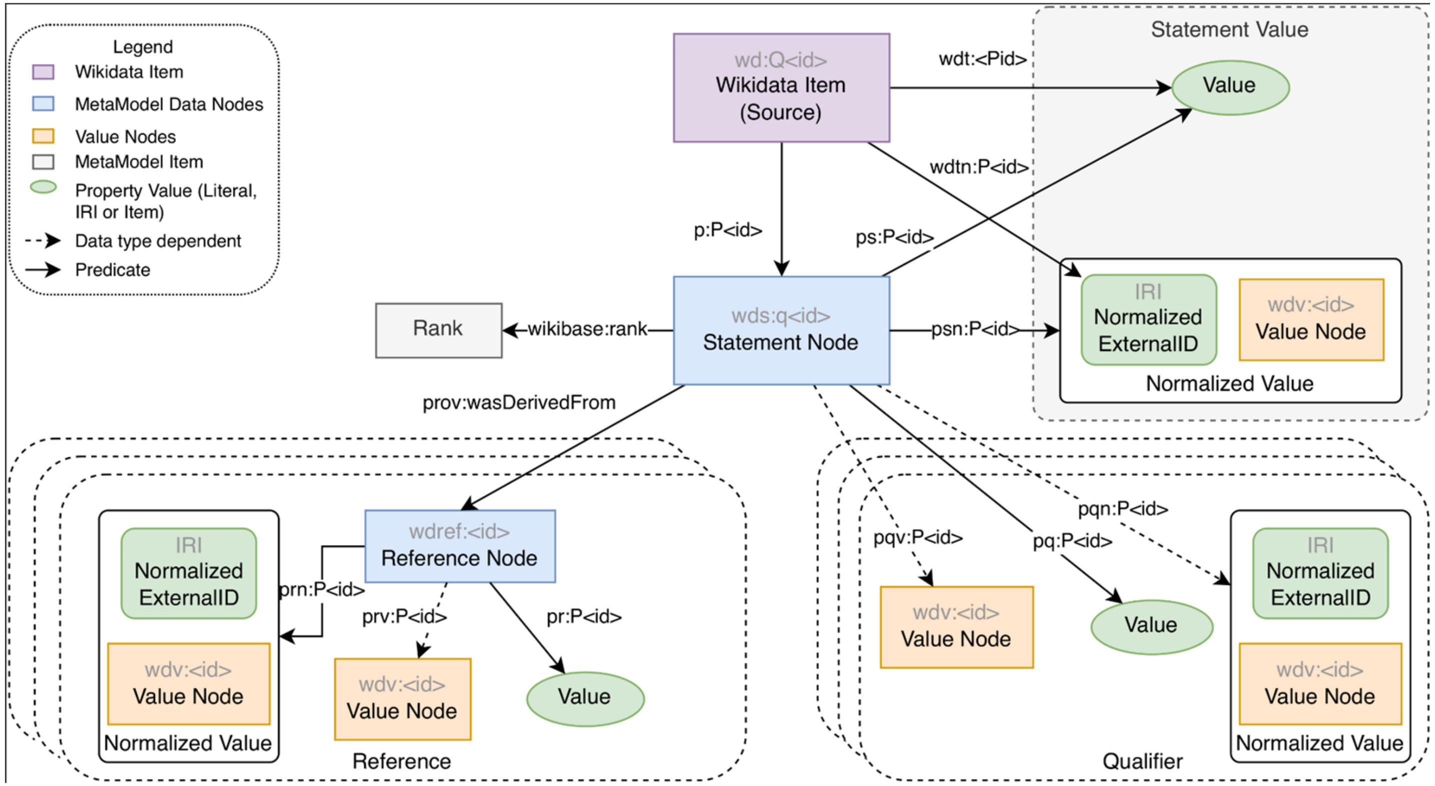
Wikidata

- One of the largest Knowledge Graphs
- Crowd-sourced and publicly available data
- Used by/as/for
  - Wikipedia
  - Referenced by other KG such as Google KG
  - Disambiguation target
  - Training LLMs
  - Question answering systems
    - such as Siri
- Covers different domains
- Data Hub for external identifiers
- Public Query and Update Endpoints





Wikidata Statement Concept

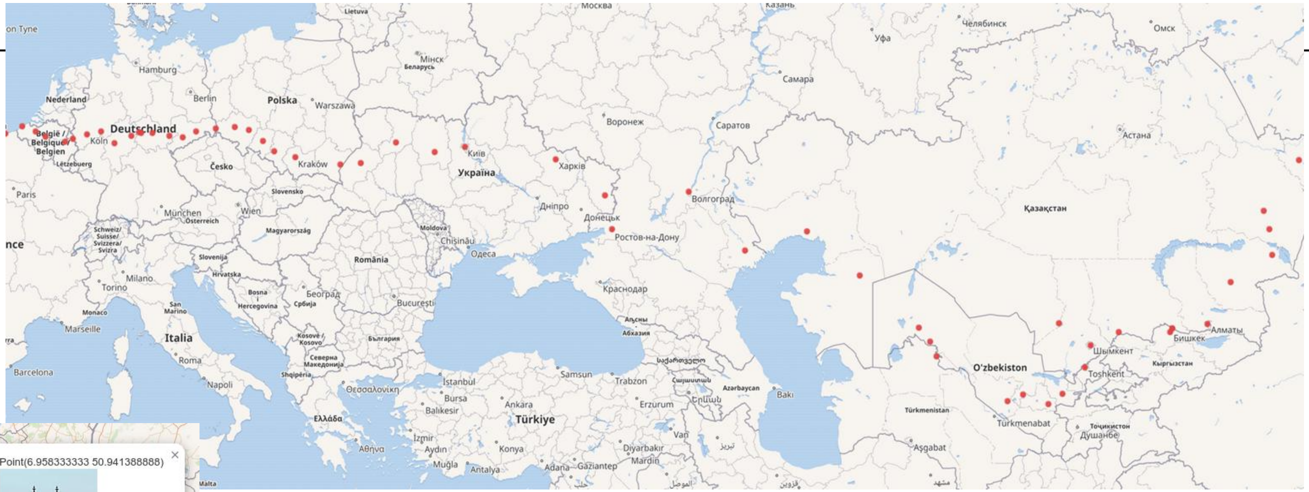


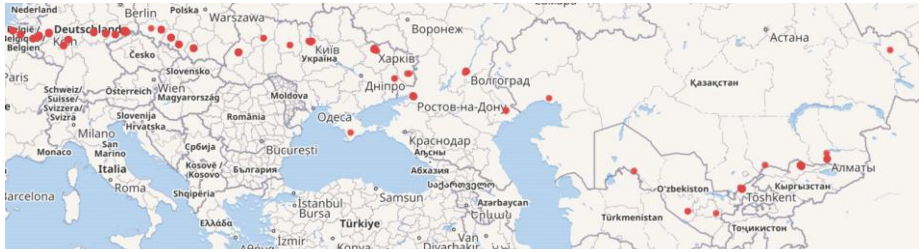
Motivation

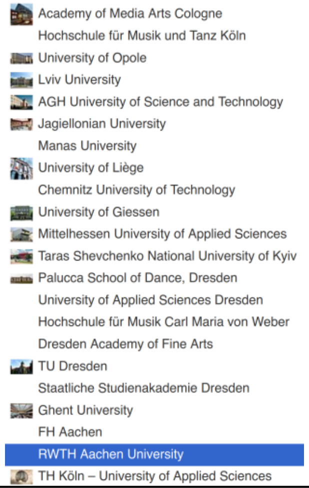
Planning a tour across europe

Traveling along the E40

- Which cities are along the way?
  - [query]
- What tourist attractions can be visited?
  - [query]
- Or visit other universities
  - [query]

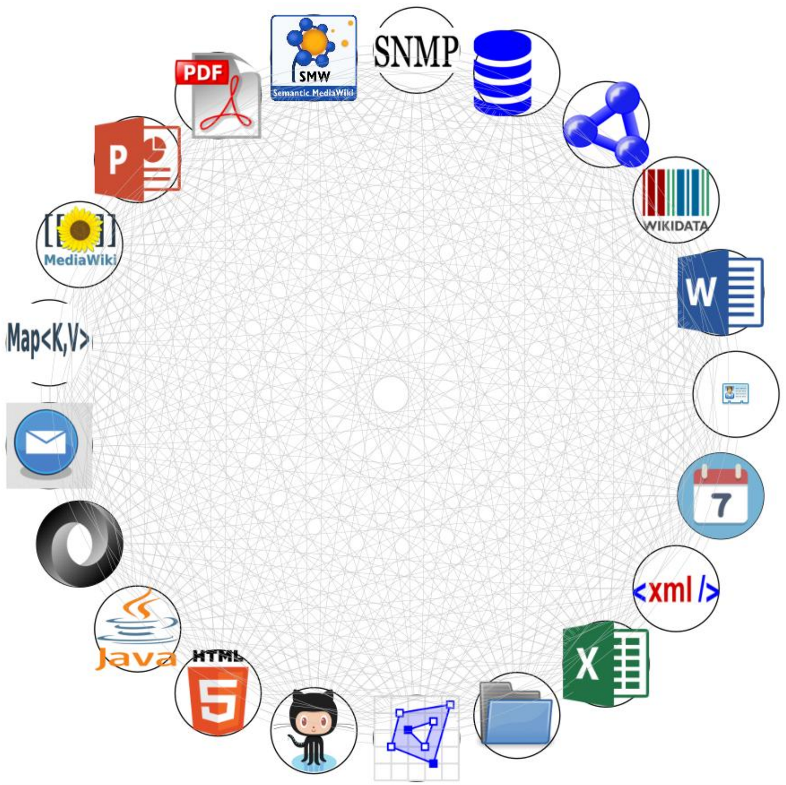






Motivation

Knowledge Graphs are useful!



See http://wiki.bitplan.com/index.php/SimpleGraph

Knowledge graphs:
- are needed for many relevant real world use-cases
- integrate and link knowledge from diverse sources
- make knowledge “gold nuggets” findable
- allow to do “look around the corner” steps quickly and many times (transitively)
- provide a basis for artificial intelligence and machine learning
- create a semantic data fabric over formerly unconnected platforms
- network effect (Metcalfe's law)

What is the KG Lab?

The knowledge graphs lab will be performed as a continuous practical exercise.
You will be given a general task description.
In your group organize, structure and outline the task and how to solve it.

The tasks will revolve around the core steps of
- Natural Language Processing (NLP)
- Named Entity Recognition (NER) and Entity Linking (NEL)
- Effective entity reconciliation
- Graph retrieval-augmented generation (graph RAG)
- Further Knowledge Graph Creation and Curation

Fraunhofer FIT
Computer Science 5 - Information Systems and Databases
RWTH AACHEN UNIVERSITY

What will we do?

Project

- Topic will be introduced today
- Time until the end of the lab (see timeline)
- Group work
- Overall workload: 6 CP (bachelor)/7 CP (master) = 180–210 h [no guarantee]
- Project meetings are scheduled within each group
- Open-source participation

What will we do?

Timeline

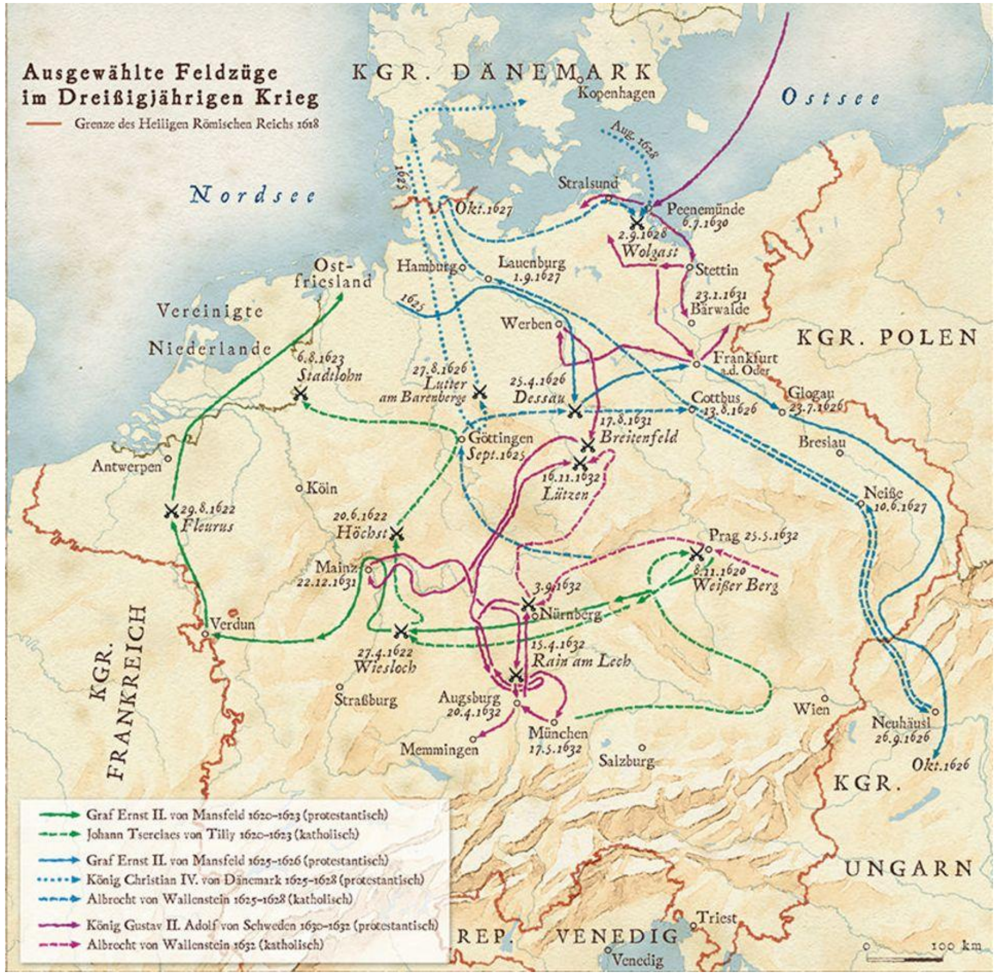
Date	Topic	Mode
2025-10-13	Kick-off, Projects and Introduction	presence
2025-10-17 23:59	End Topic Preference Selection	
2025-10-20 23:59	Deregistration Deadline	
2025-10-20	Group Inception	online
2025-10-20 – 2025-12-08	Project Phase 1	per team online
2025-12-08	Midterm Presentations	presence/online
2025-12-08 – 2026-03-11	Project Phase 2	per team online
2026-03-11 23:59	Project Result Submission	online
2026-03-15	Final Presentations	presence

Available Topics

- **Available Tasks:**
  - **Task 1 –** Interactive Temporal-Spatial Data Visualization from SPARQL
  - **Task 2 –** Context-Aware NL2SPARQL with Interactive Linking
  - **Task 3 –** Schema-Guided Data Entry for Wikibase

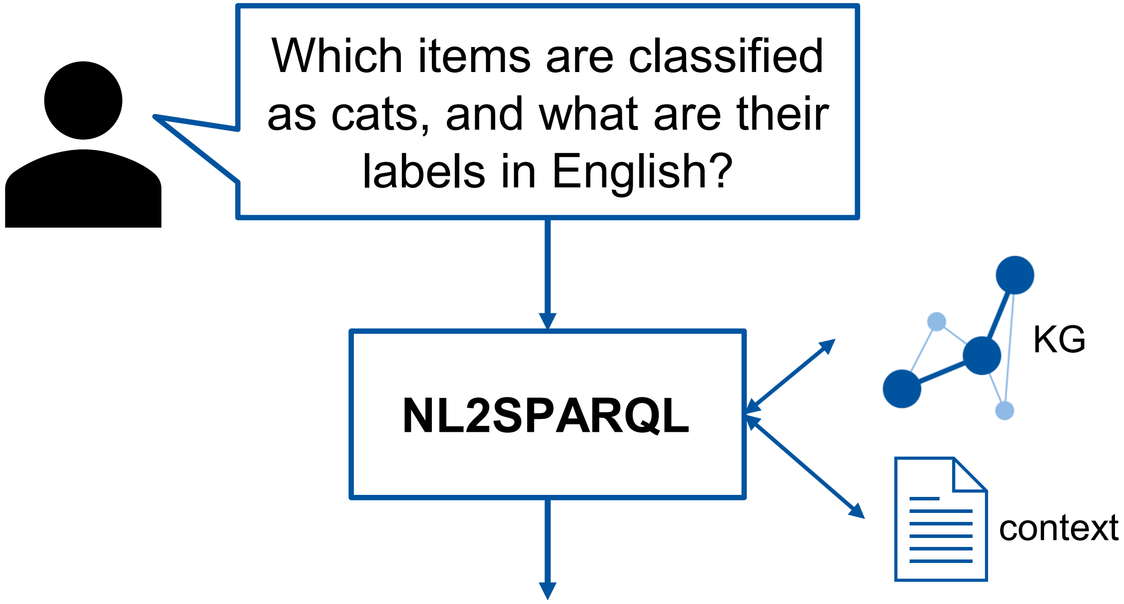
Task 1 – Interactive Temporal-Spatial Data Visualization from SPARQL

- **Goal:** Build a web-based tool that visualizes spatio-temporal data from SPARQL queries on interactive, time-aware maps
- **Motivation:** Historical and scientific datasets contain rich temporal-spatial dimensions that are difficult to understand through static tables – interactive mapping reveals patterns in movements, events, and relationships over time
- **Data Handling**: Accept SPARQL endpoints/queries with temporal ranges, locations, and movement paths using standard RDF vocabularies



Task 2 – Context-Aware NL2SPARQL with Interactive Linking

- **Goal**: Build a chatbot that converts natural language questions into SPARQL queries with configurable schema context and interactive entity-linking validation
- **Motivation**: NL2SPARQL systems often hallucinate properties and use wrong entity identifiers. Adding graph context (schemas, examples) and user-in-the-loop validation creates safer, more accurate queries for real-world knowledge graphs
- **Architecture**: Modular pipeline with mention extraction, candidate ranking, entity disambiguation, context-aware prompt construction, query generation, validation, and execution stages




```
SELECT ?item ?itemLabel
WHERE
{
  ?item wdt:P31 wd:Q146.
  ?item rdfs:label ?itemLabel. Filter(lang(?item)="en")
}

?itemLabel
```

Task 3 – Schema-Guided Data Entry for Wikibase

- **Goal**: Build a schema-driven tool that converts ShEx (EntitySchemas) into automated data entry forms for Wikibase instances (e.g. FactGrid, Wikidata)
- **Motivation**: Manual Wikibase editing is repetitive and error-prone. Schema-driven forms improve efficiency and data quality by guiding users, validating inputs in real-time, and ensuring consistent statements with qualifiers and references
- **Schema Processing:** Parse ShEx EntitySchemas into form fields reflecting cardinalities, value constraints, and support for properties, qualifiers, and references with deterministic mapping

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>

start = @<human>

<human> EXTRA wdt:P31 {
  wdt:P31 [wd:Q5];
  wdt:P18 . * ;               # image (portrait)
  wdt:P21 [wd:Q48270 wd:Q48279 wd:Q179294 wd:Q189125 wd:Q207959 wd:Q301702
wd:Q350374 wd:Q505371 wd:Q660882 wd:Q746411 wd:Q859614 wd:Q1052281 wd:Q1097630
wd:Q1289754 wd:Q1399232 wd:Q2449503 wd:Q3177577 wd:Q3277905 wd:Q6581072
wd:Q6581097 wd:Q7130936 wd:Q12964198 wd:Q15145778 wd:Q15145779 wd:Q18116794
wd:Q27679684 wd:Q27679766 wd:Q52261234 wd:Q93954933 wd:Q93955709 wd:Q96000630
wd:Q25388691 wd:Q56315990]?;   # gender
  wdt:P19 . ?;                      # place of birth
  wdt:P20 . ?;                      # place of death
  wdt:P119 . ?;                    # place of burial
  wdt:P1442 . ?;                  # image of grave
  wdt:P569 . ? ;                   # date of birth
  wdt:P570 . ? ;                   # date of death
  wdt:P735 . * ;                    # given name
  wdt:P734 . * ;                    # family name
  wdt:P106 . * ;                    # occupation
  wdt:P1559 . ? ;              #name in native language
  wdt:P27 @<country> *;           # country of citizenship
  wdt:P22 @<human> *;             # father
  wdt:P25 @<human> *;             # mother
  wdt:P3373 @<human> *;           # sibling
  wdt:P26 @<human> *;             # spouse
  wdt:P40 @<human> *;             # children
  wdt:P1038 @<human> *;           # relatives
  wdt:P103 @<language> *;         # native language
  wdt:P1412 @<language> *;        # languages spoken, written or signed
  wdt:P6886  @<language> *;       # writing language
  rdfs:label rdf:langString+;
}

<country> EXTRA wdt:P31 {
  wdt:P31 [wd:Q6256 wd:Q3024240 wd:Q3624078] +;
}

<language> EXTRA wdt:P31 {
  wdt:P31 [wd:Q34770 wd:Q1288568] +;
}
```

What are the goals?

What will you learn?

- Software development within a team
- Learning collaborative code development tools
- Learning by doing: Knowledge Graphs
  - Knowledge of how to use Wikidata and its APIs
  - Understand the Named Entity Linking (NEL) and disambiguation problem and different techniques to solve it
  - Understand the challenges and benefits of interlinking knowledge graphs
  - Understand and apply graph RAG

What are the prerequisites?

Needed base knowledge

Basic skills:
- Programming skills (**Python**)
- Query Languages (SQL/**SPARQL**/CYPHER/Gremlin …)
- Open source software development tools (GitHub/**GitLab** …)
- Data structures
- Algorithms

What is Requirements Engineering (RE)

- Structured process to *discover*, *analyze*, *specify*, *validate*, and *manage* what to build and constraints
- Benefits:
  - shared understanding,
  - reduced rework,
  - prioritized scope,
  - measurable success,
  - early risk mitigation
- Outcomes:
  - clear, testable requirements
  - in/out of scope
  - measurable quality attributes
  - assumptions and constraints

Phase 1: Workflow and Deliverables

- Understand the task:
  - unpack task and acceptance criteria
  - identify stakeholders
  - collect clarifying questions
  - review related work
- Analyze and prioritize (E.g. use MoSCoW method):
  - resolve conflicts
  - feasibility assessment
  - risk identification
- Model:
  - domain glossary
  - system context
  - key use cases/user journeys
  - data/ontology sketch
  - Interfaces
- Specify:
  - Software Requirements Specification (SRS) with acceptance criteria
  - Measurable non-functional requirements (NFRs)
  - Architecture decisions (ADRs)
  - Plan (Gantt chart) your project

Phase 1: Using acceptance criteria effectively

- Treat given criteria as scope boundaries and “definition of success”
- Decompose into user stories with acceptance criteria
- Evaluate if acceptance criteria are aligned with the goal of your project
  - It is possible to rule out acceptance criteria if a reason is provided
- Derive acceptance tests and test data
  - decide measurement methods and tools
- Add missing quality criteria:
  - security,
  - performance,
  - usability,
  - data/graph quality,
  - provenance
- **Maintain traceability and close gaps early via instructor feedback**

Phase 2: Implementation and Validation

- Follow your plan for implemantation of the project
- Acceptance- and test-driven development
  - Verify NFRs
  - run acceptance tests and demo scenarios
  - capture feedback/defects; fix and re-test.
- Definition of Done:
  - code reviewed,
  - tests passing,
  - docs updated,
  - deployment scripts working.
- AI usage:
  - allowed with verification;
  - check licenses
  - no secrets in prompts or repo

Phase 2: Implementation and Validation

- Final deliverables:
  - working system,
  - passing acceptance tests,
  - test report + metrics (CI report sufficent),
  - known limitations,
  - release notes/demo script
  - Project Paper

Final Submission an Evaluation

Final Submission

- Code (as part of open source project)
- Report (LNCS style ~8 pages)
- Final Presentation

Assessment:
- Group work (participation, quality of results) 60%
- Final written report 15%
- presentation 25%

How will you achieve the necessary knowledge to start?

Links

- KG Lab Moodle Room
- Ollama
  - Only reachable within the rwth network (use vpn)
  - Endpoint: http://ollama.warhol.informatik.rwth-aachen.de/api/chat

Papers

- Knowledge Graphs, Aidan Hogan et al, 2021
- DIN SPEC 91526:2025-05 Knowledge Graphs for Language Models and Language Models for Knowledge Graphs

Books:

- Knowledge Graphs: Methodology, Tools and Selected Use Cases, Dieter Fensel et al, Springer, 2020, ISBN 978-3030374389
- Conceptual Structures: Information Processing in Mind and Machine, John F. Sowa, Longman Higher Education, 1983, ISBN 978-0201144727

Questions/Discussion

What would you like to know?

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Next Steps

Things to do until next meeting

- Read the task descriptions thoroughly
- Enter your topic preferences in Moodle **until 2025-10-17 23:59**
- Contact your group members
- Structure the task of the work packages in issues and milestones
  - estimate how much time you need for each issue (rather small than large issues)
  - distribute the tasks by assigning the tasks
- Setup your repository in GitLab
  - README, CI, issues, milestones, etc.

Deregistration

- **if you want to deregister from the Lab you have time until next Week (2025-10-20 23:59)**
- **leaving the course afterwards will result in failing the course**

Thank you !